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views::cartesian_product

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Abstract

This paper proposes `std::ranges::cartesian_product_view` for taking the cartesian product of multiple forward ranges.

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§ 1. Changes from r0

- Specify the return type of `size`
- Add design section on the first range argument

§ 2. Motivation

`Cartesian product` is a fundamental mathematical construct. There should be a `cartesian_product_view` which generates the cartesian product of any number of ranges, e.g.:

Before	After
<pre>std::vector<int> a,b,c; for (auto&& ea : a) { for (auto&& eb : b) { for (auto&& ec : c) { use(ea,eb,ec); } } }</pre>	<pre>std::vector<int> a,b,c; for (auto&& [ea,eb,ec] : std::views::cartesian_product(a,b,c)) { use(ea,eb,ec); }</pre>

This is especially useful for composing with other views, or dealing with parameter packs of ranges:

Before	
<pre>template <std::size_t N = 0, class F, class Res, class Tuple, class... Args> auto find_tuples_satisfying_impl(F f, Res& res, Tuple const& ranges, Args const&... args) requires(N == std::tuple_size_v<std::remove_cvref_t<Tuple>>) { if (std::invoke(f, args...)) { res.push_back(std::make_tuple(args...)); } } template <std::size_t N = 0, class F, class Res, class Tuple, class... Args> auto find_tuples_satisfying_impl(F f, Res& res, Tuple const& ranges, Args const&... args) { for (auto&& e : std::get<N>(ranges)) { find_tuples_satisfying_impl<N+1>(f, res, ranges, args..., e); } } template <class F, std::ranges::forward_range... Vs> requires(std::regular_invocable< F, std::ranges::range_reference_t<Vs>...>) auto find_tuples_satisfying(F f, Vs&&... vs) { std::vector<std::tuple<std::ranges::range_value_t<Vs>...>> res; find_tuples_satisfying_impl(f, res, std::tuple(std::forward<Vs&&>(vs)...)); return res; }</pre>	<pre>template <class F, std::ranges::for requires(std::regular_invocable<F, std::ranges::range_refere auto find_tuples_satisfying(F f, Vs return std::views::cartesian_pro std::views::filter([&f](auto&& std::ranges::to<std::vector>(</pre>

§ 3. Design

§ 3.1. Minimum Range Requirements

This paper requires all ranges passed to `cartesian_product_view` to be forward ranges. See [Initial Range Argument Special-Casing](#) for discussion on the first argument being an input range.

§ 3.2. Tuple or pair

A potential design would be to use `std::tuple` as the `value_type` and `reference_type` of `cartesian_product_view`. This paper uses `std::pair` if two ranges are passed and `std::tuple` otherwise. See [p2321](#) for motivation.

§ 3.3. `reference_type`

This paper uses `tuple-or-pair<ranges::reference_t<Vs>...>` as the reference type. See [p2214](#) for discussion of value types (particularly `pair` and `tuple`) as the `reference_type` for ranges, and [p2321](#) for wording of improvements for key use-cases.

§ 3.4. Empty cartesian product view

Trying to take the cartesian product view of 0 views will produce an `empty_view<tuple<>>`, in parity with Range-v3 and [p2321](#).

§ 3.5. Common Range

`cartesian_product_view` can be a common range either if all the underlying ranges are common, or if the underlying ranges are all sized and random access. This paper reflects this.

§ 3.6. Bidirectional Range

`cartesian_product_view` can be a bidirectional range if the underlying ranges are bidirectional and common, or if they are random access and sized. Non-common bidirectional ranges are not supported because decrementing when one of the iterators is at the beginning of its range would require advancing the iterator to end, which may be linear in complexity.

We don't consider non-common, random access, sized ranges as worth supporting, so this paper requires bidirectional and common.

§ 3.7. Random Access Range

`cartesian_product_view` can be a random access range if all the underlying ranges are random access and sized. Sized ranges are required because when the view is incremented, the new states of the iterators are calculated modulo the size of their views.

We can't think of too many use-cases for this and it adds a fair bit of implementation burden, but this paper supports the necessary operations.

§ 3.8. Initial Range Argument Special-Casing

The first range passed to `cartesian_product_view` can be treated specially, since it is only passed through a single time. Therefore, the specification could relax several constraints on the first range passed:

- It could be an input range instead of forward range
- It wouldn't need to be a sized range in order for the `cartesian_product_view` to be random-access or common

- It wouldn't need to be common in order for the `cartesian_product_view` to be bidirectional

However, we don't consider these use-cases to be strong enough for the inconsistency which they would bring.

§ 3.9. Sized Range

`cartesian_product_view` can be a sized range if all the underlying ranges are, in which case the size is the product of all underlying sizes. This is reflected in the paper.

§ 3.10. Naming

An alternative name is `std::ranges::product_view`. This paper uses `cartesian_product_view` as we believe it is more explicit in what its semantics are.

§ 3.11. Pipe Support

It may be possible to support syntax such as `vec1 | views::cartesian_product(vec2)` by either fixing the number of arguments allowed to the view, or adding a pipe operator to `cartesian_product_view` itself.

However, it's problematic for the same reason as `views::zip`, in that one cannot distinguish between the usages in `a | views::cartesian_product(b, c)` and `views::cartesian_product(b, c)` on its own. As such, this paper does not support this syntax.

§ 4. Implementation

There are implementations of a cartesian product view in `Range-v3`, `cor3ntin::rangesnext`, and `tl::ranges`, among others.

§ 5. Wording

§ 5.1. Cartesian Product view [range.cartesian]

§ 5.1.1. Overview [range.cartesian.overview]

`cartesian_product_view` presents a `view` with a value type that represents the cartesian product of the adapted ranges.

The name `views::cartesian_product` denotes a customization point object. Given a pack of subexpressions `Es...`, the expression `views::cartesian_product(Es...)` is expression-equivalent to

- `*decay-copy*(views::empty<tuple<>>)` if `Es` is an empty pack,
- otherwise, `cartesian_product_view<views::all_t<decltype((Es))>...>(Es...)`.

[Example:

```
std::vector<int> v { 0, 1, 2 };
for (auto&& [a,b,c] : std::views::cartesian_product(v, v, v)) {
    std::cout << a << ' ' << b << ' ' << c << '\n';
    //0 0 0
    //0 0 1
    //0 0 2
    //0 1 0
    //0 1 1
    //...
}
```

-- end example]

§ 5.1.2. Class template `cartesian_product_view` [range.cartesian.view]

```

namespace std::ranges {
    template <class... Vs>
    concept cartesian-product-is-random-access = //exposition only
        ((random_access_range<Vs> && ...) &&
         (sized_range<Vs> && ...));

    template <class... Vs>
    concept cartesian-product-is-bidirectional = //exposition only
        ((bidirectional_range<Vs> && ...) && (common_range<Vs> && ...));

    template <class... Vs>
    concept cartesian-product-is-common = //exposition only
        (std::ranges::common_range<Vs> && ...)
        || cartesian-product-is-random-access<Vs...>;

    template<class... Ts>
    using tuple-or-pair = decltype([]{
        // exposition only
        if constexpr (sizeof...(Ts) == 2) {
            return type_identity<pair<Ts...>>{};
        }
        else {
            return type_identity<tuple<Ts...>>{};
        }
    }())::type;

    template<class F, class Tuple>
    constexpr auto tuple-transform(F&& f, Tuple&& tuple) // exposition only
    {
        return apply([&]<class... Ts>(Ts&&... elements){
            return tuple-or-pair<invoke_result_t<F&, Ts>...>{
                invoke(f, std::forward<Ts>(elements))...
            };
        }, std::forward<Tuple>(tuple));
    }

    template<class F, class Tuple>
    constexpr void tuple-for-each(F&& f, Tuple&& tuple) // exposition only
    {
        apply([&]<class... Ts>(Ts&&... elements){
            (invoke(f, std::forward<Ts>(elements)), ...);
        }, std::forward<Tuple>(tuple));
    }

    template<forward_range... Vs>
    requires (view<Vs> && ...)
    class cartesian_product_view
    : public view_interface<cartesian_product_view<Vs...>> {
    private:
        std::tuple<Vs...> bases_; //exposition only

        template<bool Const>
        struct iterator; //exposition only
        template<bool Const>
        struct sentinel; //exposition only
    public:
        constexpr cartesian_product_view() = default;
        constexpr cartesian_product_view(Vs... bases);

        constexpr iterator<false> begin() requires(!simple_view<Vs> || ...);
        constexpr iterator<true> begin() const requires (range<const Vs> && ...);

        constexpr iterator<false> end()

```

```

requires ((!simple_view<Vs> || ...) && cartesian-product-is-common<Vs...>);
constexpr iterator<true> end() const
requires(cartesian-product-is-common<const Vs...>);
constexpr default_sentinel_t end() const
requires (!cartesian-product-is-common<const Vs...>);

constexpr *see below* size() requires (sized_range<Vs> && ...);
constexpr *see below* size() const requires (sized_range<const Vs> && ...);
};

template <class... Vs>
cartesian_product_view(Vs&&...) -> cartesian_product_view<all_t<Vs>...>;

namespace views { inline constexpr unspecified cartesian_product = unspecified; }
}

```

```
constexpr cartesian_product_view(Vs... bases);
```

Effects: Initialises `bases_` with `std::move(bases)...`

```
constexpr iterator<false> begin() requires(!simple_view<Vs> || ...);
```

Effects: Equivalent to `return iterator<false>(tuple-transform(ranges::begin, bases_));`

```
constexpr iterator<true> begin() requires (range<Vs> && ...);
```

Effects: Equivalent to `return iterator<true>(tuple-transform(ranges::begin, bases_));`

```
constexpr iterator<false> end()
requires ((!simple_view<Vs> || ...) && cartesian-product-is-common<Vs...>);
```

Effects: If every range in `Vs` models `random_access_range`, equivalent to `return begin() + size()`. Otherwise equivalent to:

```

iterator<false> it(tuple-transform(ranges::begin, bases_));
std::get<0>(it.current_) = std::ranges::end(std::get<0>(bases_));
return it;

```

```
constexpr iterator<true> end() const
requires(cartesian-product-is-common<const Vs...>);
```

Effects: If every range in `Vs` models `random_access_range`, equivalent to `return begin() + size()`. Otherwise equivalent to:

```

iterator<true> it(tuple-transform(ranges::begin, bases_));
std::get<0>(it.current_) = std::ranges::end(std::get<0>(bases_));
return it;

```

```
constexpr default_sentinel_t end() const
requires (!cartesian-product-is-common<const Vs...>) {
```

Effects: Equivalent to `return default_sentinel;`.

```
constexpr *see below* size() requires (sized_range<Vs> && ...);
constexpr *see below* size() const requires (sized_range<const Vs> && ...);
```

The return type is the smallest unsigned integral type that is sufficiently wide to store the product of the maximum sizes of all underlying ranges. If there is no such type, the type is implementation-defined.

Effects: Returns the product of the size of all ranges in `bases_`.

5.1.3. Class template `cartesian_product_view::iterator` [ranges.cartesian.iterator]

```
namespace std::ranges {
template<forward_range... Vs>
requires (view<Vs> && ...)
template<bool Const>
class cartesian_product_view<Vs...>::iterator {
    maybe-const<Const, cartesian_product_view*> parent_; //exposition only
    tuple-or-pair<iterator_t<maybe-const<Const, Vs>>...> current_{}; //exposition only

    template <size_t N = (sizeof...(Vs) - 1)>
    void next(); //exposition only

    template <size_t N = (sizeof...(Vs) - 1)>
    void prev(); //exposition only

public:
    using iterator_category = input_iterator_tag;
    using iterator_concept = *see below*;
    using value_type = tuple-or-pair<range_value_t<maybe-const<Const, Vs>>...>;
    using difference_type = std::common_type_t<range_difference_t<maybe-const<Const, Vs>>...>;

    iterator() = default;
    constexpr explicit iterator(tuple-or-pair<iterator_t<maybe-const<Const, Vs>>...> current_):
        current_(current_) {}

    constexpr iterator(iterator<!Const> i) requires Const && (std::convertible_to<
        std::ranges::iterator_t<Vs>,
        std::ranges::iterator_t<maybe-const<Const, Vs>>> && ...);

    constexpr auto operator*() const;
    constexpr iterator& operator++();
    constexpr iterator operator++(int);

    constexpr iterator& operator--();
    requires (cartesian-product-is-bidirectional<maybe-const<Const, Vs>...>);
    constexpr iterator operator--(int);
    requires (cartesian-product-is-bidirectional<maybe-const<Const, Vs>...>);

    constexpr iterator& operator+=(difference_type x);
    requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);
    constexpr iterator& operator-=(difference_type x);
    requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);

    constexpr reference operator[](difference_type n) const;
    requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);

    friend constexpr bool operator==(const iterator& x, const iterator& y)
    requires (equality_comparable<iterator_t<maybe-const<Const, Vs>>> && ...);

    friend constexpr bool operator==(const iterator& x, const std::default_sentinel_t&);
```

```

friend constexpr auto operator<(const iterator& x, const iterator& y)
requires (std::ranges::random_access_range<maybe-const<Const, Vs>> && ...);
friend constexpr auto operator>(const iterator& x, const iterator& y)
requires (std::ranges::random_access_range<maybe-const<Const, Vs>> && ...);
friend constexpr auto operator<=(const iterator& x, const iterator& y)
requires (std::ranges::random_access_range<maybe-const<Const, Vs>> && ...);
friend constexpr auto operator>=(const iterator& x, const iterator& y)
requires (std::ranges::random_access_range<maybe-const<Const, Vs>> && ...);

friend constexpr auto operator<=>(const iterator& x, const iterator& y)
requires ((std::ranges::random_access_range<maybe-const<Const, Vs>> && ...) &&
          (std::ranges::three_way_comparable<maybe-const<Const, Vs>> && ...));

friend constexpr iterator operator+(const iterator& x, difference_type y)
requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);
friend constexpr iterator operator+(difference_type x, const iterator& y)
requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);
friend constexpr iterator operator-(const iterator& x, difference_type y)
requires (cartesian-product-is-random-access<maybe-const<Const, Vs>...>);
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
requires (sized_sentinel_for<iterator_t<maybe-const<Const, Vs>>, iterator_t<maybe-const
}
}

```

`iterator::iterator_concept` is defined as follows:

- If every type in `Vs` models `random_access_range`, then `iterator_concept` denotes `random_access_iterator_tag`.
- Otherwise, if every type in `Vs` models `bidirectional_range`, then `iterator_concept` denotes `bidirectional_iterator_tag`.
- Otherwise, `iterator_concept` denotes `forward_iterator_tag`.

```

template <size_t N = (sizeof...(Vs) - 1)>
void next() //exposition only

```

Effects: Equivalent to:

```

auto& it = get<N>(current_);
++it;
if constexpr (N > 0) {
    if (it == ranges::end(get<N>(parent_->bases_))) {
        it = ranges::begin(get<N>(parent_->bases_));
        next<N - 1>();
    }
}

```

```

template <size_t N = (sizeof...(Vs) - 1)>
void prev() //exposition only

```

Effects: Equivalent to:

```
auto& it = std::get<N>(current_);
if (it == std::ranges::begin(std::get<N>(parent_>bases_))) {
    std::ranges::advance(it, std::ranges::end(std::get<N>(parent_>bases_)));
    if constexpr (N > 0) {
        prev<N - 1>();
    }
}
--it;
```

```
constexpr explicit iterator(tuple-or-pair<iterator_t<maybe-const<Const, Vs>>...> current);
```

Effects: Initializes current_ with std::move(current).

```
constexpr iterator(iterator<!Const> i)
    requires Const && (convertible_to<iterator_t<Vs>, iterator_t<maybe-const<Const, Vs>>> &&
```

Effects: Initializes current_ with std::move(i.current_).

```
constexpr auto operator*() const;
```

Effects: Equivalent to:

```
return tuple-transform([](auto& i) -> decltype(auto) { return *i; }, current_);
```

```
constexpr iterator& operator++();
```

Effects: Equivalent to:

```
next();
return *this;
```

```
constexpr iterator operator++(int);
```

Effects: Equivalent to:

```
auto tmp = *this;
++*this;
return tmp;
```

```
constexpr iterator& operator--() requires (bidirectional_range<maybe-const<Const, Vs>> &&
```

Effects: Equivalent to:

```
prev();  
return *this;
```

```
constexpr iterator operator--(int) requires (bidirectional_range<maybe-const<Const, Vs>> &&
```

Effects: Equivalent to:

```
auto tmp = *this;  
--*this;  
return tmp;
```

```
constexpr iterator& operator+=(difference_type x) requires (random_access_range<maybe-const
```

Effects: Sets the position of the iterators in `current_`:

- If $x > 0$, as if `next` was called x times.
- Otherwise, if $x < 0$, as if `prev` was called $-x$ times.
- Otherwise, no effect.

```
constexpr iterator& operator-=(difference_type x) requires (random_access_range<maybe-const
```

Effects: Equivalent to:

```
*this += -x;  
return *this;
```

```
constexpr reference operator[](difference_type n) const requires (random_access_range<maybe
```

Effects: Equivalent to `return *((*this) + n);`.

```
friend constexpr bool operator==(const iterator& x, const iterator& y)  
requires (equality_comparable<iterator_t<maybe-const<Const, Vs>>> && ...);
```

Effects: Equivalent to `return x.current_ == y.current_;`.

```
friend constexpr auto operator<(const iterator& x, const iterator& y)  
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to `return x.current_ < y.current_;`.

```
friend constexpr auto operator>(const iterator& x, const iterator& y)  
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `y < x;`.

```
friend constexpr auto operator<=(const iterator& x, const iterator& y)
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `!(y < x);`.

```
friend constexpr auto operator>=(const iterator& x, const iterator& y)
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `!(x < y);`.

```
friend constexpr auto operator<=>(const iterator& x, const iterator& y)
requires (random_access_range<maybe-const<Const, Vs>>&&...) && (three_way_comparable<maybe-c
```

Effects: Equivalent to return `x.current_ <=> y.current_;`.

```
friend constexpr iterator operator+(const iterator& x, difference_type y)
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `iterator{ x } += y;`.

```
friend constexpr iterator operator+(difference_type x, const iterator& y)
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `y + x;`.

```
friend constexpr iterator operator-(const iterator& x, difference_type y)
requires (random_access_range<maybe-const<Const, Vs>> && ...);
```

Effects: Equivalent to return `iterator{ x } -= y;`.

```
friend constexpr difference_type operator-(const iterator& x, const iterator& y)
requires (sized_sentinel_for<iterator_t<maybe-const<Const, Vs>>, iterator_t<maybe-const<Con
```

Effects: Let `scaled_distance(x,y,N)` be the distance between `(get<N>(x.current_) - get<N>(y.current_)) * get<N>(x.parent_->bases_)`. Returns the sum of `scaled_distance(x,y,N)` for every `N` in the interval `(0, sizeof...(Vs)]`.

§ 5.1.4. Class template `cartesian_product_view::sentinel` [ranges.cartesian.sentinel]

```
template <bool Const>
class sentinel {
    using parent = maybe-const<Const, cartesian_product_view>; //exposition only
    using first_base = decltype(get<0>(std::declval<parent>()).bases_); //exposition only
    sentinel_t<first_base> end_; //exposition only

public:
    sentinel() = default;
```

```

    sentinel(std::ranges::sentinel_t<first_base> end);

    friend constexpr bool operator==(sentinel const& s, iterator_t<parent> const& it);

    constexpr sentinel(sentinel<!Const> other)
    requires Const &&
        (std::convertible_to<std::ranges::sentinel_t<first_base>, std::ranges::sentinel_t<
};

```

```

sentinel(std::ranges::sentinel_t<first_base> end);

```

Effects: Initialises `end_` with `std::move(end)`.

```

constexpr sentinel(sentinel<!Const> other)
requires Const &&
    (std::convertible_to<std::ranges::sentinel_t<first_base>, std::ranges::sentinel_t<const
    : end_(other.end_) {}

```

Effects: Initialises `end_` with `std::move(other.end_)`.

```

friend constexpr bool operator==(sentinel const& s, iterator_t<parent> const& it);

```

Effects: Equivalent to `return get<0>(it.current_) == s.end_;`.

§ 6. Acknowledgements

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